

The Role of Gravitational Perturbations in Determining Jupiter's Mass: A Comparative Study of Io and Callisto

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Kepler's third law provides a fundamental approach to estimating a planet's mass. It assumed an idealised system where perturbations are neglected, and celestial bodies are treated as point masses. This paper discusses how perturbations influence the accuracy of Jupiter's mass estimation determined using Kepler's third law by tracking the orbital motions of Io, its closest Galilean moon, and Callisto, its furthest. It was hypothesised that Callisto would yield a more accurate estimate, as it experiences weaker perturbative effects. Multiple telescope images of both moons were analysed using Gaia, and their distances from Jupiter were modelled to determine their semi-major axes and orbital periods. Using Kepler's third law, Jupiter's mass was estimated as $(2.0 \pm 0.5) \times 10^{27}$ kg from Io and $(1.91 \pm 0.03) \times 10^{27}$ kg from Callisto, compared to the accepted value of 1.898×10^{27} kg. The analysis of perturbations, including solar influence, mutual satellite interactions, and Jupiter's oblateness, showed that Io is more affected by Jupiter's oblateness than Callisto. This supports the hypothesis that Callisto provides a more accurate mass estimate as it is less susceptible to perturbative influences.

I. INTRODUCTION

Jupiter, the largest of the gas giants, is the most massive planet in our solar system. It orbits the Sun at an average distance of 5.20 AU [8] and is situated near other gas giants, including Saturn, Uranus, and Neptune. With the second-highest number of natural satellites [10], Jupiter's four largest and most massive moons — Io, Europa, Ganymede, and Callisto — are collectively known as the Galilean moons [9].

Determining Jupiter's mass is essential for space exploration, as precise calculations are required for spacecraft and probe trajectories, particularly for gravity assist manoeuvres around the planet. However, precisely measuring its mass using the orbits of the Galilean moons is challenging due to various perturbative effects.

The primary method to determine the mass of a planet with moons is to use Kepler's third law [12], which is expressed as

$$T^2 = \frac{4\pi^2 a^3}{G(M_1 + M_2)} \quad (1)$$

where T , a and M_1 are the orbital periods, semi-major axes and masses of its moons respectively and M_2 is the mass of the planet. Since the masses of Io and Callisto which are used in this paper are negligibly small compared to that of Jupiter, the M_1 term can be ignored, and equation (1) is simplified to

$$T^2 = \frac{4\pi^2 a^3}{GM_J} \quad (2)$$

where M_J is the mass of Jupiter. Rearranging equation (2) yields

$$M_J = \frac{4\pi^2 a^3}{GT^2} \quad (3)$$

Kepler's third law is derived from Newton's law of universal gravitation and the centripetal force acting on a body orbiting a spherically symmetric planet. The law assumes an idealised two-body system in which celestial bodies are treated as point masses [11], meaning their mass is effectively concentrated at their centres. However, in reality, deviations arise due to planetary oblateness caused by

rotation, solar perturbations, mutual interactions between moons, and gravitational influences from other celestial bodies. These factors can affect the mass estimation of a planet derived from Kepler's laws.

This paper describes an investigation which hypothesises that Callisto, the most distant of Jupiter's Galilean moons, will yield a more precise estimate of Jupiter's mass than Io, its closest Galilean moon. Since Callisto orbits farther from Jupiter, it is less affected by Jupiter's oblateness and potential perturbations from orbital resonance. The latter was initially considered to affect only Callisto, as it is not part of the Laplace resonance (orbital resonance with three celestial bodies), which involves Io, Europa, and Ganymede with integer-related orbital periods of 4:2:1 [3].

II. METHODOLOGY

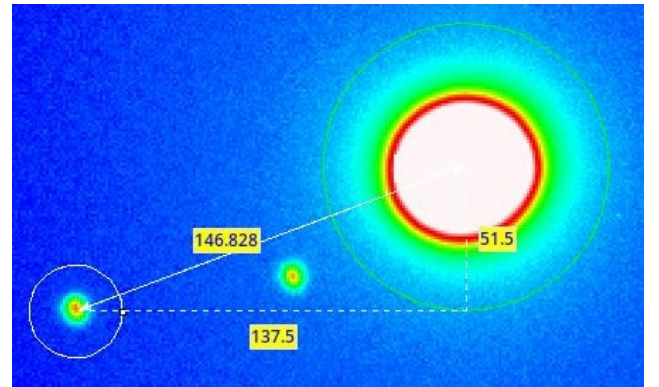


FIG. 1: The position of Io was determined using the StarLink website [6]. The colour map was set to 'Colour,' and the colour scale was adjusted to 'square root' for improved visualisation. Circles were drawn around Jupiter and its moons for a more precise estimation of their centres.

A telescope with a plate scale of 0.983 arcsecond per pixel [1] was used to capture multiple images of Jupiter and its Galilean moons over a two-week period to track the positions of Io and Callisto. To accurately identify Io and Callisto among the Galilean moons, an external website [6] was used. The images were processed using Gaia. The linear distance from Jupiter to each moon was measured in pixels

by constructing a right triangle, where the hypotenuse extends from the centre of Jupiter to the respective moon, as shown in FIG. 1.

The measured distances (in pixels), d were converted to angular separations, θ by multiplying d with 4.77×10^{-6} rad. The separations, L between Jupiter and the moons were then calculated using the equation

$$L = r\theta \quad (4)$$

where r is the real-time distance from Earth to Jupiter, which is obtained from [6] and small angle approximation was used.

Graphs of separation distances between Jupiter and its moons over time (in days) were plotted, assuming a model of the form

$$f(t) = A \sin(Bt + C) + D \quad (5)$$

The semi-major axes, A and orbital periods, $2\pi/B$ were obtained from the best-fit parameters and substituted into equation (3) to estimate the mass of Jupiter.

III. RESULTS

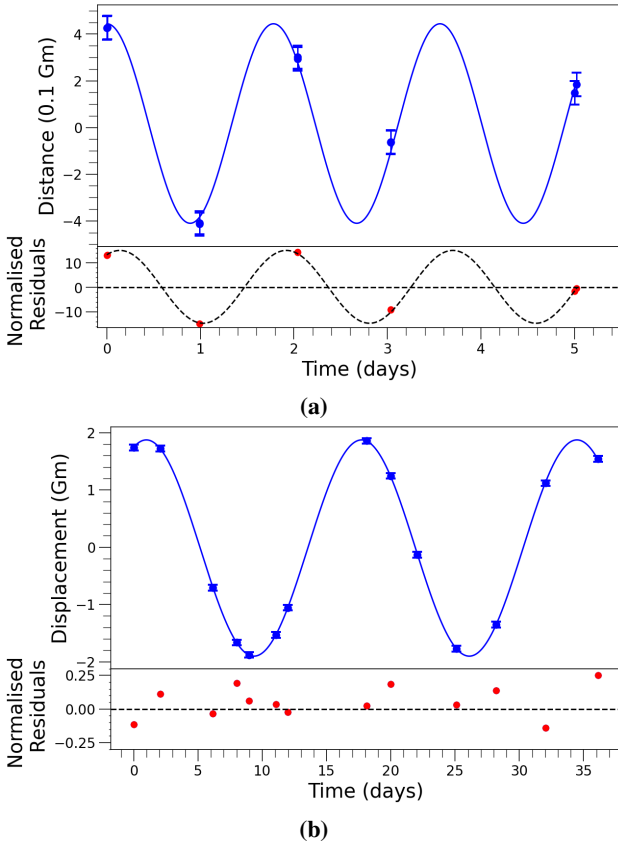


FIG. 2: (a) Displacement-time graph of Io, with potential pattern in the residual plot (b) Displacement-time graph of Callisto

From FIG. 2(a), the orbital period of Io is determined to be (1.78 ± 0.03) days, while for Callisto, FIG. 2(b) gives an orbital period of (16.76 ± 0.04) days. The semi-major axes, obtained from the amplitudes of the graphs, are calculated as $(4.3 \pm 0.3) \times 10^8$ m for Io and $(1.89 \pm 0.01) \times 10^9$ m for Callisto. These values are in agreement with [9].

Substituting these values into equation (3), along with the gravitational constant $G = 6.67 \times 10^{-11} \text{ m}^3 / \text{kg s}^2$, yields Jupiter's mass as $M_J^{(I)} = (2.0 \pm 0.5) \times 10^{27} \text{ kg}$ when using Io's data and $M_J^{(C)} = (1.91 \pm 0.03) \times 10^{27} \text{ kg}$ when using Callisto's data.

FIG. 2(a) shows a reduced chi-squared value of χ_v^2 of 0.46 with a p-value of $P(\chi_{min}^2, v) = 0.84$ where v is degree of freedom. Compared to an ideal fit, where $\chi_v^2 \approx 1$ and $P(\chi_{min}^2, v) \approx 0.5$, this suggests that the errors are either overestimated or the model is overfitting the data. A potential periodic pattern is observed, which may be due to an insufficient number of data points or the presence of periodic perturbations. The later sections will quantify and determine whether this pattern arises from data insufficiency or periodic perturbations.

FIG. 2(b) shows a reduced chi-squared value of $\chi_v^2 = 0.082$ and a p-value of $P(\chi_{min}^2, v) = 0.99$, indicating that the model is overfitting the data and the errors are overestimated. Multiple attempts were made to refine the model and improve these values, with the current model yielding the best results. This suggests the need for further investigation into error estimates and model complexity.

Aside from the random errors arising from the measurement of distances between the centres of Jupiter and its moons, the uncertainties in the calculated values are also influenced by the interactions and dynamics of the planetary system. These factors will be further explored in the following section.

IV. ANALYSIS AND MODELLING PERTURBATIONS

A. Solar Perturbation

Solar perturbation was initially thought to have a significant impact on the calculation of Jupiter's mass due to the Sun's massive gravitational influence. To quantify this, the Hamiltonian H_{Sun} describing the interactions between Io and Callisto under the Sun's perturbative effects is obtained from [3]

$$H_{Sun} = -\frac{Gm_i m_\odot}{a_\odot} \left\{ \frac{1}{2} b_{1/2}^{(0)} \left(\frac{a_i}{a_\odot} \right) - 1 + \frac{1}{8} \frac{a_i}{a_\odot} b_{3/2}^{(1)} \left(\frac{a_i}{a_\odot} \right) (e_i^2 + e_\odot^2) \right\} \quad (6)$$

Here, the subscript i indicates that the orbital elements are associated with either Io or Callisto. The eccentricities e_i and $e_\odot$ represent the orbital eccentricities of Io or Callisto and Jupiter's orbit around the Sun, respectively. The terms $b_s^{(j)}(\alpha)$ where $\alpha = a_i/a_{Sun}$, denote Laplace coefficients [3], defined as

$$b_s^{(j)}(\alpha) = \frac{1}{\pi} \int_0^{2\pi} \frac{\cos(j\psi) d\psi}{(1 - 2\alpha \cos \psi + \alpha^2)^s} \quad (7)$$

Since $\alpha \ll 1$, the equation (7) is approximated using a binomial series expansion up to the second degree, yielding

$$b_{1/2}^{(0)}(\alpha) = 2 + \frac{1}{2}\alpha^2 \quad (8)$$

$$b_{3/2}^{(1)}(\alpha) = 3\alpha \quad (9)$$

Only terms up to the second degree in α are considered, as higher-order terms are negligible.

To quantify the effect of solar perturbation, we consider the time derivative of the canonical momentum in Hamilton's equations, $\dot{p} = -\partial H/\partial q$, which corresponds to the force acting on the celestial bodies. Since the only relevant force is gravitational, and solar perturbation remains within the Newtonian limit, differentiating equation (6) with respect to a_i yields the perturbative force exerted by the Sun on Io and Callisto:

$$F_{\text{perturb}}^{(Sun)} = \frac{Gm_i m_s}{2a_{\odot}^2} \left[\frac{a_i}{a_{\odot}^2} + \frac{3a_i(e_i + e_{\odot})}{8} \left(\frac{1}{a_{\odot}^2} + \frac{1}{a_{\odot}} \right) \right] \quad (10)$$

The total force, F_T , acting on Io or Callisto is then the sum of the gravitational attraction from Jupiter and the perturbative force from the Sun:

$$F_T = \frac{M_J m_i}{a_i^2} + F_{\text{perturb}}^{(Sun)} \quad (11)$$

Setting equation (11) equal to the required centripetal force and solving for Jupiter's mass yields the modified mass estimate

$$\tilde{M}_J = \frac{4\pi^2 a_i^3}{GT^2} - \frac{m_{\odot} a_i^2}{2a_{\odot}^2} \left[\frac{a_i}{a_{\odot}^2} + \frac{3a_i(e_i + e_{\odot})}{8} \left(\frac{1}{a_{\odot}^2} + \frac{1}{a_{\odot}} \right) \right] \quad (12)$$

By substituting the semi-major axes and masses of Io and Callisto and the parameters given in [2], the perturbative influence of the Sun on the calculation of Jupiter's mass can be evaluated. The results indicate that the perturbative effect is significantly smaller than 1%, implying that the impact of solar perturbation on Jupiter's mass determination is negligible.

B. Mutual Satellites' Interactions

The Jovian system, or Jupiter's planetary system, consists of 95 satellites [8], all of which gravitationally interact with one another. Among them, the Galilean moons exert the most significant gravitational influence due to their larger masses. The perturbation effects caused by these moons can be evaluated using a similar approach to the one outlined in the previous section.

The Hamiltonian for the direct part of the disturbing function is given by:

$$H_{\text{int}} = -\frac{Gm_i m_{Eu}}{|a_{Eu} - a_i|} - \frac{Gm_i m_{Gan}}{|a_{Gan} - a_i|} - \frac{Gm_i m_j}{|a_i - a_j|} \quad (13)$$

where m_{Eu} and a_{Eu} are the mass and semi-major axis of Europa, and m_{Gan} and a_{Gan} correspond to that of Ganymede. The subscripts i and j represent the elements associated to Io or Callisto. The coordinate system is centred at Jupiter's centre of mass for simplicity, allowing the indirect term to be ignored. Differentiating equation (13) with respect to a_i gives the perturbative force acting on Io

or Callisto:

$$F_{\text{perturb}}^{(\text{satellite})} = \frac{Gm_i m_{Eu}}{|a_i - a_{Eu}|^3} (a_i - a_{Eu}) + \frac{Gm_i m_{Gan}}{|a_i - a_{Gan}|^3} (a_i - a_{Gan}) + \frac{Gm_i m_j}{|a_j - a_i|^3} (a_j - a_i) \quad (14)$$

The total force acting on Io or Callisto is thus the sum of Jupiter's gravitational attraction and the perturbative forces exerted by the other Galilean moons. This modifies the estimated mass of Jupiter, expressed as:

$$M'_J = \frac{4\pi^2 a_i^3}{GT^2} - \frac{a_i^2 m_{Eu}}{|a_i - a_{Eu}|^3} (a_i - a_{Eu}) - \frac{a_i^2 m_{Gan}}{|a_i - a_{Gan}|^3} (a_i - a_{Gan}) - \frac{a_i^2 m_j}{|a_j - a_i|^3} (a_j - a_i) \quad (15)$$

By substituting the relevant parameters into equation (15) and converting the result into percentage form, the perturbative effect can be quantified. For Io, the perturbation introduces a 0.01% deviation in the mass estimate, whereas for Callisto, the effect is 0.06%. This suggests that Callisto's mass estimate is more affected by the gravitational perturbations of the other Galilean moons than Io's.

C. Jupiter's Oblateness

Jupiter has the fastest rotation speed in the solar system, completing one rotation in just 9.93 hours [8]. This rapid spin causes the planet to bulge at the equator, resulting in a deviation from a perfect spheroidal shape. Consequently, Jupiter's mass distribution is not uniform, meaning it no longer strictly adheres to the assumptions of Newton's law of universal gravitation. This oblateness introduces perturbations [7] to its potential energy, ϕ , which can be approximated using series expansion.

To simplify the calculation, Jupiter is assumed to have a constant density throughout the volume and exhibit axial symmetry. The simplified potential energy, evaluated at the equatorial plane [13], is expressed as

$$\tilde{\phi}(a_i) = -\frac{GM_J m_i}{a_i} \left[1 - \sum_{n=1}^{\infty} (-1)^n \frac{(2n)!}{(2^n n!)^2} J_{2n} \left(\frac{R}{a_i} \right)^{2n} \right] \quad (10)$$

where J_{2n} is the dimensionless multipole moment coefficient of Jupiter and R is the mean equatorial radius. The subscript i retains its usual meaning, as defined in the discussion previously. Since J_2 is the largest term [13], the series in equation (10) is truncated at the first degree, yielding

$$\tilde{\phi}(a_i) = -\frac{GM_1 M_2}{a_i} \left[1 + J_2 \left(\frac{R}{a_i} \right)^2 \right] \quad (16)$$

By taking the derivative of equation (16), the gravitational force acting on Io and Callisto is expressed as

$$F(r) = \frac{d\tilde{\phi}}{da_i} = \frac{GM_J m_i}{a_i^2} \left(1 + \frac{3J_2 R^2}{a_i^2} \right) \quad (17)$$

Equating equation (17) with the centripetal force, the modified mass of Jupiter, $\tilde{M}_J$, is

$$\tilde{M}_J = \left(\frac{a_i^2}{a_i^2 + 3J_2 R^2} \right) M_J \quad (18)$$

Using $J_2 = 0.014736$ for Jupiter [8], the deviations in its mass due to oblateness are found to be approximately 0.12% for Io and 0.0063% for Callisto, based on their respective semi-major axes. However, this value represents a lower bound for the perturbation, as Jupiter has a density gradient and is not perfectly axially symmetric. Factors such as its asymmetric magnetosphere [5] and atmospheric disturbances — most notably the Great Red Spot, which is active in the southern hemisphere — further contribute to deviations from the simplified model.

V. DISCUSSION

The perturbation caused by other planets is neglected in this report, as it primarily induces long-term effects [7] on the orbits of Io and Callisto rather than the short-term effects that are the focus of this study. Consequently, the dominant perturbative effects considered are solar perturbation, mutual interactions between the satellites, and Jupiter's oblateness.

As analysed in the previous section, solar perturbation affects Io and Callisto similarly, despite the Sun being the most massive body in the Solar System. While mutual satellite interactions exert a stronger perturbative effect on Io than on Callisto, this difference is significantly smaller than the effect of Jupiter's oblateness. This indicates that Jupiter's oblateness is the dominant perturbation influencing both moons' orbits. Given their proximity to Jupiter and its immense mass, this result is expected. Io, being the closest Galilean moon to Jupiter, experiences the strongest perturbation due to its oblateness. Moreover, this difference suggests that the potential sine-wave pattern observed in the residuals of Io's displacement-time graph is at least partially attributed to Jupiter's oblateness, though further data points are needed for confirmation.

This analysis considers each perturbation separately while neglecting others. A more comprehensive approach would involve solving an n-body problem [7] and simulations, which is beyond the scope of this report. However, more precise values for perturbations could be obtained by extending the summation to higher-order terms in equation (10).

VI. CONCLUSION

The calculations show that using the masses and semi-major axes of Io and Callisto yields different results: $(2.0 \pm 0.5) \times 10^{27}$ kg for Io and $(1.91 \pm 0.03) \times 10^{27}$ kg for Callisto. Compared to NASA's reported value of 1.898×10^{27} kg [8], Callisto provides a more accurate estimate, as it experiences less perturbation from Jupiter's oblateness than Io. Therefore, the hypothesis is accepted.

VII. REFERENCES

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VIII. ERROR APPENDIX

This section is written with reference to Measurement and their Uncertainties by Hughes and Hase [4].

A. Error Bars

The uncertainties in the distances between Jupiter and its moons (y error bars) arise from determining their centres. FIG. 1 was cropped to isolate Jupiter, Io, and Callisto for error analysis. To remove unnecessary colours, the colour map was set to greyscale, and the colour scale was reverted to its default setting, rendering the images in grey. The images were then imported into a Python script to extract the pixel intensities in both horizontal and vertical directions, and 3D Gaussian plots were generated to calculate the standard deviations along the x-axis, σ_x and y-axis, σ_y .

A discrepancy was found between the actual size of the image and its size when opened by an image viewer. When the images were sent to the Python script, the number of pixels read by the script was scaled up, leading to deviated standard deviations. To correct this, another Python script was run to determine the number of pixels in the scaled-up images, and the real number of pixels was manually counted. The scaling factor was calculated by dividing the number of pixels in the scaled-up images by the real number of pixels.

The image of Jupiter was over-exposed, creating a flat surface on the 3D Gaussian, as shown in FIG. 3. Hence, the 3D Gaussian was extrapolated from the tail to the peak to obtain accurate standard deviations.

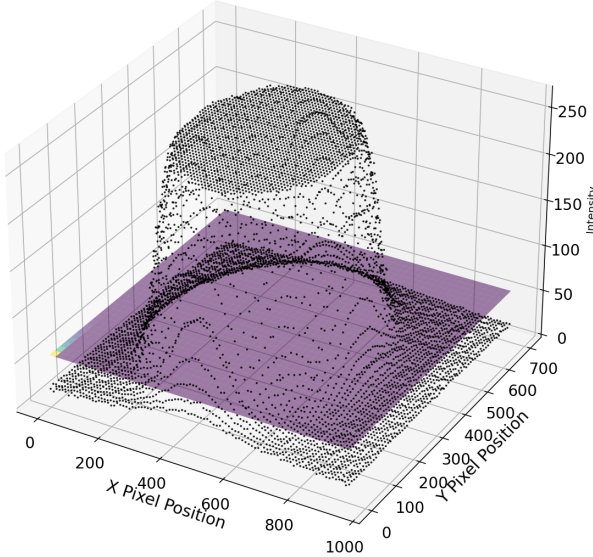


FIG. 3: The CCD of the telescope was saturated, leading to a constant intensity, which resulted in an incomplete 3D Gaussian model.

The final standard deviations, σ_f in pixels were calculated as the quadrature of σ_x and σ_y

$$\sigma_f = \sqrt{\sigma_x^2 + \sigma_y^2} \quad (19)$$

The final standard deviations are listed below:

Jupiter/moons	σ_x (pixels)	σ_y (pixels)	σ_f (pixels)
Jupiter	12.7	9.42	15.8
Io	2.25	1.89	2.94
Callisto	2.32	2.05	3.10

TABLE I: Table of σ_x , σ_y and σ_f of Jupiter and its moons

Similar to the conversion and equation (4) suggested in Methodology section, the y-error bars are calculated as

$$\alpha = \frac{0.983(\sigma_f^{(J)} + \sigma_f^{(i)})r}{206265} \quad (20)$$

where r and superscript i have the usual meanings.

B. Uncertainties in Orbital Periods and Semi-major Axes

The uncertainties in the best-fit parameters are taken to be the standard errors, α , which can be obtained by taking the square root of the diagonal elements of the covariance matrix, C

$$\alpha_j = \sqrt{C_{jj}} \quad (21)$$

where α_j is uncertainty of the j-th parameter, and C is the inverse of curvature matrix, A , defined by

$$A_{jk} = \frac{1}{2} \frac{\partial^2 \chi^2}{\partial a_j \partial a_k} \quad (22)$$

where a_j and a_k are the j-th and k-th parameter, respectively.

Standard errors of parameters A and B in equation (5) correspond to the uncertainties in the semi-major axes of the moons and the orbital speed of their orbits, respectively. The uncertainties in the orbital period and mass of Jupiter can be obtained through standard error propagation.

IX. LAY SUMMARY FOR GENERAL AUDIENCE

Jupiter is the largest and most massive planet in our solar system. It has a complex system of 95 moons, with four of them—Io, Europa, Ganymede, and Callisto—being the most massive. These four moons are known as the Galilean moons, discovered by Galileo Galilei in 1610 using his self-built telescope.

One way to estimate Jupiter's mass is by studying the Galilean moons, using equation (3). However, this equation doesn't take into account external factors, such as the Sun's gravity, the mutual effects between the moons, and oblateness of Jupiter (shown in FIG.4). This report explores how these factors, or perturbations, affect the accuracy of estimating Jupiter's mass by tracking the positions of Io, the closest moon, and Callisto, the furthest.

Telescope images of Io and Callisto were analysed to obtain their distances from Jupiter, a and time taken for them to complete an orbit, T . The data is then substituted in equation (3) to find the mass estimate of Jupiter. As expected, Callisto provided a more accurate result than Io, as it is less affected by the overall perturbation.

Understanding Jupiter's mass and how its moons are influenced by these factors is crucial for space exploration. Accurate calculations are needed for planning spacecraft and probe missions, especially for gravity assist manoeuvres (shown in FIG.5).

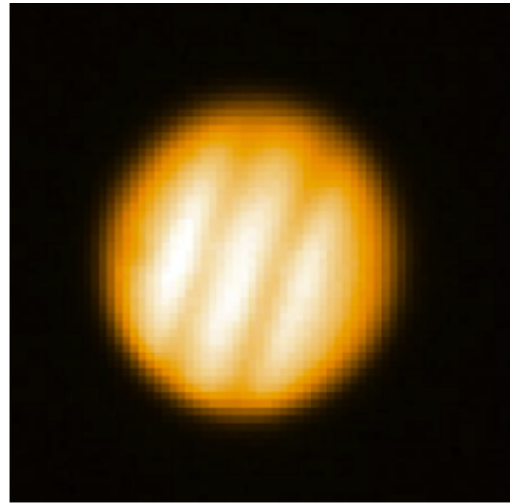


FIG. 4: Due to Jupiter's rapid rotation, its equator appears bulged, making it an oblate spheroid rather than a perfect sphere.

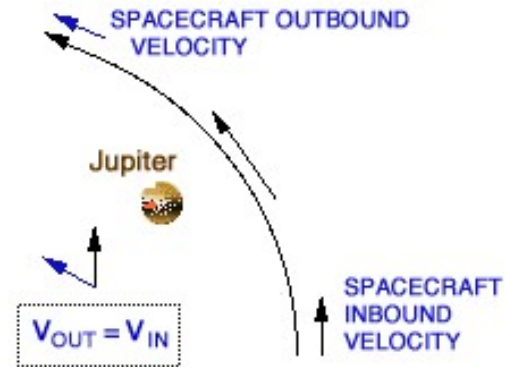


FIG. 5: The spacecraft approaches Jupiter and is redirected at a higher speed. It is a manoeuvre that helps accelerate and conserve fuel. Credit: The Worlds of David Darling

X. GENERATIVE AI STATEMENT

In the production of this scientific report, ChatGPT was used to correct grammar, improve the coherence of sentences, fix spelling mistakes, and enhance the flow of passages. The work and content are entirely my own, and the reference sources were selected without reliance on ChatGPT.